EMBEDDED vs VLSI

The Domain Guide No One Gave You in College

Which domain to choose · What to learn · Where the jobs are

India's semiconductor industry is growing at **26% CAGR**.
Tata, Micron, CG Power — the hiring wave is real.
Which domain puts **YOU** in that wave?

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01 / WHAT ARE THEY?

EMBEDDED SYSTEMS	VLSI DESIGN
Software that runs *on top of* hardware. You write firmware, drivers, and RTOS-based code that controls microcontrollers and processors. Think: the code inside your smartwatch, car ECU, or IoT sensor.	Designing the hardware *itself* — at transistor and gate level. You design chips: RTL coding, synthesis, physical design, verification, DFT. Think: the team that designed the chip your smartwatch runs on.

Embedded → **talks to the chip**

VLSI → **IS the chip**

02 / WHAT YOU NEED TO LEARN

EMBEDDED SYSTEMS

Domain	Topics	Level
Core Language	C / C++ for embedded	**Must**
Microcontrollers	ARM Cortex-M, AVR, STM32, ESP32	**Must**
RTOS	FreeRTOS, Zephyr	**Must**
Protocols	UART, SPI, I2C, CAN, USB	**Must**
Linux Embedded	Device Drivers, BSP, Yocto	**Intermediate**
Debugging	JTAG, Logic Analyzer, Oscilloscope	**Must**
Assembly	ARM Assembly basics	**Good to have**

VLSI DESIGN

Domain	Topics	Level
RTL Design	Verilog / SystemVerilog	**Must**
Verification	UVM, SystemVerilog Assertions	**Must**
Synthesis	Design Compiler, Genus	**Must**
Physical Design	Floorplan, P&R, STA, SignOff	**Must**
DFT	Scan, ATPG, MBIST, Boundary Scan	**Must**
EDA Tools	Synopsys, Cadence, Mentor	**Must**
Analog Basics	CMOS, transistor-level design	**Good to have**

03 / WHICH DOMAIN SUITS YOU?

Be honest with yourself. Pick based on what excites you — not what sounds fancy.

If you…	**Go for →**
Love writing code and seeing hardware react	**Embedded**
Want to understand how chips are made	**VLSI**
Enjoy debugging hardware with oscilloscopes	**Embedded**
Are fascinated by logic gates, timing, silicon	**VLSI**
Want to work in automotive / IoT / robotics	**Embedded**
Want to work at Intel, Qualcomm, AMD, Apple	**VLSI**
Prefer software-heavy roles	**Embedded**
Love deep engineering at transistor level	**VLSI**
College had good micro/processor labs	**Embedded**
College had VLSI / EDA tool exposure	**VLSI**

■ **Pro tip:** You don't have to pick just one forever. Many senior engineers understand both. Start with one, go deep, then expand.

04 / WHERE THE JOBS ARE

● EMBEDDED — Top Hiring Companies

Company	Roles	Fresher CTC
Bosch	Embedded C, AUTOSAR, CAN	5–8 LPA
Continental	Automotive Embedded, RTOS	5–9 LPA
Texas Instruments	Firmware, DSP, MCU	8–14 LPA
STMicroelectronics	Embedded SW, Drivers	6–10 LPA
Qualcomm	Embedded Linux, BSP	10–18 LPA
Tata Elxsi	ADAS, Embedded Systems	4–7 LPA
Wipro / HCL	IoT, Embedded projects	3.5–6 LPA

● VLSI — Top Hiring Companies

Company	Roles	Fresher CTC
Intel	RTL Design, Physical Design, DFT	12–22 LPA
Qualcomm	RTL, Verification, DFT	14–24 LPA
Synopsys	EDA Tools, DFT, STA	10–18 LPA
Cadence	PD, Verification, AMS	10–18 LPA
Samsung Semi	SOC Design, Memory Design	10–20 LPA
MediaTek	SOC Design, DFT, Synthesis	10–18 LPA
Tata / Micron (India)	Process, DFT, PD	8–16 LPA

* CTC ranges are indicative. Actual offers vary by company, role, and interview performance.

05 / START FOR FREE

● Embedded — Free Tools

→ Arduino IDE → easiest entry point for microcontrollers
→ PlatformIO → professional embedded dev (VS Code plugin)
→ STM32CubeIDE → official IDE for STM32 boards
→ QEMU → emulate ARM hardware without physical board
→ Wokwi → online Arduino/ESP32 simulator, zero hardware needed
→ FreeRTOS → open source RTOS, widely used in industry

● VLSI — Free Tools

→ EDA Playground → browser-based Verilog/SystemVerilog simulator
→ Icarus Verilog + GTKWave → open source simulation & waveform viewer
→ Yosys → open source synthesis tool
→ OpenROAD → full open source RTL-to-GDS physical design flow
→ Vivado WebPACK → Xilinx FPGA tool, free edition
→ OpenLane → automated RTL-to-GDS2 flow (Google MPW program)

The honest truth?

Your college name won't get you shortlisted. Your **skills** will.

Pick a domain. Go deep. Build projects. Stay consistent.
India's semiconductor wave is just beginning — and there is space
for engineers who actually know their stuff.

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